

NVH Source Locator — Quick Reference

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A one-page refresher. For full details, see **User Guide**.

Core flow (2-Sensor, free)

- **Pick a material** — Materials tab → tap your material
- **Enter calibration** on the 2-Sensor tab: - Sensor spacing (d) - Calibration time delay (tCa l) — auto-filled from material
- **Enter event** — tEvent and First sensor (A or B)
- **Read result** — distance from sensor A



NVH Source Locator
 TDOA Calculator PRO




2-Sensor 3-Sensor 3-Sen+ 4-Sensor 4-Sen+ 3D 3D+ Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor tab

All tabs

Tab	Output	Pro fields?
2-Sensor	Distance along line	No (fully free)
3-Sensor	X, Y on a surface	Yes
3-Sen+	X, Y with LSQ over 3 pairs	Yes

4-Sensor	X, Y from two pairs (A–B + C–D)	Yes
4-Sen+	X, Y from 4 sensors, any position	Yes
3D	X, Y, Z from 4 sensors	Yes
3D+	X, Y, Z from up to 6 sensors	Yes
Materials	Speed-of-sound picker	No
Help	Tutorials	No

Settings is the  icon (top-right), not a tab.

Temperature compensation

Settings → Reference temperature, range **-40 to +200 °C**.

- **14 metals** have built-in compensation (aluminium, steels, copper, brass, bronze, titanium, magnesium, lead, zinc, nickel, tungsten, iron, iron cast)
- Materials without compensation show **"ref only"**
- **Resets to 20 °C on every app launch** (default-safe-start)
- Replaying a history entry restores its original temperature

Shortcuts

- **Tap a material** → auto-fills all `tCaL` fields across all tabs
- **Hold +/-** on number fields → fast increment
- **Drag horizontally** on a number field → scrub values
- **Empty/negative/garbage input** → snaps to 0 on blur (temp input clamps to -40/200)
- **Star a material** → moves to top of picker

Pro model

Feature-locked freemium (\$19.99):

- Free: 2-Sensor tab fully functional, no limits
- Pro: Other tabs accessible but have **gold-padlock fields** that show paywall on tap

Pro unlocks: 3-Sensor through 3D+, custom materials, backup/restore, PDF reports, photo annotation.



Unlock NVH Pro

Full access to all features

- ✓ 3-Sensor & 4-Sensor triangulation
- ✓ 3D source location (X, Y, Z)
- ✓ Unlimited measurement history
- ✓ Save custom materials
- ✓ Backup and restore
- ✓ PDF reports with photo annotation
- ✓ Temperature compensation across all tabs

Unlock Pro — \$19.99

Have a Google Play promo code?

Looking for a discount code?
[Check the Picoscope Forum →](#)

Questions? support@evdiag.net

Paywall

Reports & Backup

Print result button on any result screen → PDF with header, inputs, result, visualization, photo (if taken), and temperature footer (when compensation active).

Customize header in Settings → Report header.

Backup: Settings → Backup → share to cloud/email.

Restore: Settings → Restore → pick backup file.

Restore Pro on a new device

Same Google account (Android) or Apple ID (iOS) you bought with → Settings → **Restore purchase** → unlocks within seconds.

Auto-restore happens silently when you return to the app after redeeming a promo code externally.

Quick troubleshooting

- **Result outside range?** Check tEvent sign / First sensor / sensor spacing
- **Closest material wrong?** Reference temperature probably accidentally set — check Settings
- **Restore purchase fails?** Verify same store account; reinstall if it persists
- **Field snapped to 0?** Empty/negative inputs auto-snap on blur — re-enter the value
- **Stepper buttons gone?** They appear next to fields with data-step — restart app if missing
- **Stale temperature warning?** It resets to 20 every launch — set again for this session

Contact support@evdiag.net — include device model, app version (Settings → bottom), and a description of what you tried.